

Source Heterogeneity, Not Algorithm Choice, Drives Performance Variation in Ames Mutagenicity QSAR: A Multi-Level Evaluation Across Three Genotoxicity Endpoints

Seungmin Yoon, Sanghun Kim, and Hyunpyo Jeon*

Department of Biosafety, Kyungsung University, Busan, Republic of Korea

E-mail: hpjeon@ks.ac.kr

Abstract

Quantitative structure–activity relationship (QSAR) models for Ames mutagenicity are increasingly used in ICH M7 assessments,¹ yet reported performance varies greatly depending on evaluation methodology. Here we present a systematic evaluation of 375 model configurations across five genotoxicity datasets—Ames mutagenicity ($n = 13,560$), in vitro chromosome aberration ($n = 1,619$), in vivo micronucleus ($n = 2,153$), and two negative-sampled variants—using seven algorithms, five feature representations, and three preprocessing scenarios. Using a single locked configuration (XGBoost, compact descriptors, unprocessed SMILES) evaluated at three levels of increasing stringency, we show that for Ames, performance decreased from random-split cross-validation ($\text{MCC} = 0.670$) to scaffold-split cross-validation (0.624 , $\Delta = 0.046$) to cross-source leave-domain-out in the drug-to-industrial direction (0.234). Source heterogeneity—arising from the mixing of drug-like compounds (positive rate 58%) and industrial chemicals (positive rate 3%), which may differ in data provenance, chemical

space, and labeling conventions—drove a scaffold-to-LDO MCC drop of $\Delta = 0.390$, substantially exceeding the split-method effect. A naive source classifier achieved MCC = 0.608 without any structural information. However, algorithm robustness under domain shift was not uniform: Random Forest maintained MCC = 0.325 in the most challenging cross-domain direction where XGBoost collapsed to 0.122. In vivo micronucleus prediction showed some ranking ability (ROC-AUC = 0.860, PR-AUC = 0.237) after salt stripping but lacked reliable classification utility (MCC 95% CI crossing zero). We propose multi-level evaluation with source-awareness as a reporting standard for Ames mutagenicity QSAR and provide an open-source pipeline for reproducibility.

Acknowledgement

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Introduction

Regulatory Context

The ICH M7(R2) guideline recommends the use of two complementary in silico approaches for assessing the mutagenic potential of pharmaceutical impurities: a rule-based expert system and a statistical QSAR model.¹ This regulatory framework has driven considerable interest in developing predictive models for the Ames bacterial reverse mutation assay,^{2,3} with extensions to other genotoxicity endpoints including in vitro chromosome aberration (CA) and in vivo micronucleus (MN) tests.^{4,5} Reliable QSAR models for these endpoints could reduce animal testing, accelerate safety assessments, and support the 3Rs principle.⁶

The Overestimation Problem

Published Ames QSAR models report a wide range of performance metrics, with accuracies typically spanning 0.80–0.90 and MCC values of 0.60–0.80.^{7–10} However, these numbers arise from different data splits, datasets, preprocessing pipelines, and evaluation protocols, making direct comparison difficult.

Several factors can inflate apparent model performance. First, random data splitting allows structurally similar compounds (sharing the same Murcko scaffold¹¹) to appear in both training and test sets, creating an optimistic estimate of generalization to novel structures. Scaffold-based splitting has been recommended as a more realistic evaluation strategy,^{12,13} but the magnitude of performance difference between random and scaffold splits has not been systematically quantified across multiple genotoxicity endpoints.

Second, most large Ames datasets aggregate compounds from distinct sources—typically pharmaceutical compound libraries and industrial chemical registries.^{7,9} These sources differ in composition, mutagenicity prevalence, and potentially in data quality and labeling conventions. When these heterogeneous sources are mixed in training and test sets, a model may partly learn source membership rather than toxicity mechanisms—a form of source heterogeneity that has been noted^{14,15} but, to our knowledge, never systematically measured.

Third, preprocessing choices such as salt stripping, metal compound removal, and charge normalization can substantially alter molecular representations,^{16,17} yet these steps are often undocumented or applied uniformly without assessing endpoint-specific effects.

Gaps in the Current Literature

Most published studies evaluate a single endpoint using a single split method and focus their analysis on algorithm comparison.^{18,19} Cross-domain generalization is rarely tested. Statistical significance of model differences is seldom reported.²⁰ The relative contributions of split method, source composition, and preprocessing choices to apparent model performance have not been disentangled within a single consistent experimental framework.

Contributions

This work makes the following contributions:

1. We introduce a three-level evaluation framework that progressively increases evaluation rigor—in-domain scaffold split, random-versus-scaffold comparison, and cross-source leave-domain-out—applied to a single locked model configuration for fair comparison.
2. We demonstrate that, for the Ames mutagenicity benchmark studied here, source heterogeneity arising from the mixing of drug-like and industrial compounds with a 19.5-fold prevalence difference drove a scaffold-to-LDO MCC drop of 0.390, substantially exceeding the split-method effect ($\Delta = 0.046$).
3. We show that while in-domain algorithm differences were small across all seven algorithms (MCC range 0.063 under a fixed feature/preprocessing configuration), cross-domain robustness varied meaningfully—Random Forest maintained $\text{MCC} = 0.325$ under domain shift where XGBoost collapsed to 0.122.
4. We demonstrate that in vivo micronucleus models showed some ranking signal ($\text{ROC-AUC} = 0.860$, $\text{PR-AUC} = 0.237$) after salt stripping but lacked reliable classification utility, with MCC confidence intervals crossing zero.
5. We demonstrate endpoint-specific preprocessing effects, with salt stripping improving in vivo best-model MCC by +0.22 while having mixed or negative effects on other endpoints.
6. We provide an open-source pipeline with all experimental configurations, split seeds, and locked results tables for full reproducibility.

Materials and Methods

Datasets

Five genotoxicity datasets were evaluated (Table 1). The three primary datasets comprise Ames bacterial reverse mutation ($n = 13,560$ after conflict resolution), in vitro chromosome aberration ($n = 1,619$), and in vivo micronucleus ($n = 2,153$). Two additional negative-sampled variants (in vitro sampling, $n = 345$; in vivo sampling, $n = 231$) were evaluated as exploratory supplementary analyses with reduced negative compounds.

The Ames dataset was compiled from pharmaceutical (drug-like) and industrial chemical sources. Compounds were annotated with source labels based on their originating data source. We use the term “source” to emphasize that this annotation reflects data provenance—encompassing potential differences in chemical space, data quality, and labeling conventions—rather than a computed chemical classification. Drug-sourced compounds ($n = 6,416$) had a positive rate of 58.3%, while industrial-sourced compounds ($n = 7,144$) had a positive rate of 3.0%, a 19.5-fold difference. In vitro and in vivo datasets lacked source annotations.

Label conflicts (compounds with contradictory labels across sources) were resolved conservatively by exclusion: 169 for Ames, 3 for in vitro, and 2 for in vivo.

Cross-endpoint overlap analysis revealed substantial compound sharing: 97.5% of in vitro compounds and 92.1% of in vivo compounds appeared in the Ames dataset, with moderate label concordance (Cohen’s $\kappa = 0.14$ – 0.20). This overlap indicates that the three endpoints largely probe the same compound universe with different biological readouts, a point we return to in the Discussion.

All datasets were split into training (80%) and test (20%) partitions using scaffold-based splitting¹² with zero scaffold group overlap. Murcko scaffolds¹¹ were generated using RDKit,²¹ with acyclic compounds clustered by agglomerative clustering on Morgan fingerprint²² similarity.

Table 1: Dataset summary. Positive rate refers to the training partition. The 375 total configurations comprise 5 datasets $\times$ 3 preprocessing scenarios $\times$ (6 tabular models $\times$ 4 feature modes + 1 GNN).

Dataset	Assay	n	Train	Test	Pos. rate (%)	Scaffolds
Ames	Bacterial reverse mutation	13,560	10,848	2,712	29.1	1,691
In vitro CA	Chromosome aberration	1,619	1,295	324	10.4	353
In vivo MN	Micronucleus	2,153	1,722	431	3.7 / 2.8 ^a	410
In vitro CA (sampled) ^b	Neg. sampled CA	345	276	69	49.6	169
In vivo MN (sampled) ^b	Neg. sampled MN	231	184	47	33.3	129

^aTraining / test positive rate. ^bSupplementary exploratory datasets.

Preprocessing Scenarios

Three preprocessing scenarios were applied independently to each dataset:

Raw (raw_all): Original SMILES used without modification.

Salt-stripped (salt_stripped): The largest molecular fragment was retained using RDKit’s `LargestFragmentChooser`,²¹ followed by charge normalization. This modified 45% of Ames, 81% of in vitro, and 83% of in vivo SMILES.

Metal-removed (no_metal): Compounds containing metal atoms were excluded.

Feature Representations

Five feature representations were evaluated:

Compact (138 features): 13 physicochemical descriptors (MW, LogP, TPSA, HBD, HBA, rotatable bonds, fraction C_{sp3}, aromatic ring count, ring count, heteroatom count, heavy atom count, formal charge, valence electrons) and 125 structural alert features based on Benigni–Bossa SA1–SA31,^{23,24} Kazius toxicophores,²⁵ and ISS Ames alerts.

Broad FP-512/1024/2048 (266/394/522 features): Compact features plus 128/256/384 Morgan fingerprint bits²² selected by mutual information²⁶ on the training set only. Bits with prevalence below 1% or above 99% were excluded before MI ranking.

Graph: Molecular graphs with 9-dimensional atom features and normalized adjacency matrices, used exclusively with the GCN model.

Algorithms

Seven classification algorithms were evaluated: XGBoost²⁷ and LightGBM²⁸ (gradient boosting), Random Forest²⁹ (bagging ensemble), SVM with RBF kernel³⁰ and Logistic Regression³¹ (both with standard scaling), MLP³² (128–64 hidden units, early stopping), and a three-layer Graph Convolutional Network³³ with learnable weights, LayerNorm, and global mean pooling. Class imbalance was addressed using `scale_pos_weight` or `class_weight='balanced'`. Hyperparameter tuning was performed using `RandomizedSearchCV` with `GroupKFold` (scaffold groups).³⁴

Three-Level Evaluation Framework

To ensure fair comparison across evaluation levels, we defined a locked reference configuration: XGBoost with compact features and raw SMILES. All three levels were evaluated using this single configuration:

Level 1—In-domain scaffold split: Trained on the scaffold-disjoint training partition, evaluated on the held-out test partition. MCC³⁵ with 500-iteration bootstrap 95% CIs, plus ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, and Brier score.³⁶

Level 2—Random versus scaffold split comparison: Both 5-fold stratified random CV (5 repeats) and 5-fold scaffold-grouped CV (5 repeats) on the full dataset.

Level 3—Cross-source leave-domain-out (LDO): For Ames, trained on one source and tested on the other, using the locked configuration plus two additional algorithms (LightGBM, Random Forest) to assess algorithm-specific robustness under source shift.

Additional Analyses

Full grid evaluation: Beyond the locked configuration, all 375 combinations were evaluated on the scaffold-split test set. The 375 total comprises 5 datasets $\times$ 3 preprocessing scenarios $\times$ (6 tabular algorithms $\times$ 4 feature modes + 1 GNN) = $5 \times 3 \times 25 = 375$.

Applicability domain (AD): Tanimoto similarity on binary 1024-bit Morgan fingerprints.²² Threshold ≥ 0.3 defined AD membership.

Feature importance: SHAP values³⁷ (**TreeExplainer**) for tree-based models on the three primary endpoints.

Learning curves: Training set fractions 10%–100% with 5-fold scaffold CV, using the locked configuration. Learning curves for alternative configurations (e.g., salt-stripped broad FP) were not computed and represent a limitation.

Statistical comparison: McNemar’s test³⁸ with continuity correction for pairwise comparison of the top two models per endpoint.

Threshold tuning: Out-of-fold probability-based threshold optimization using 5-fold GroupKFold.

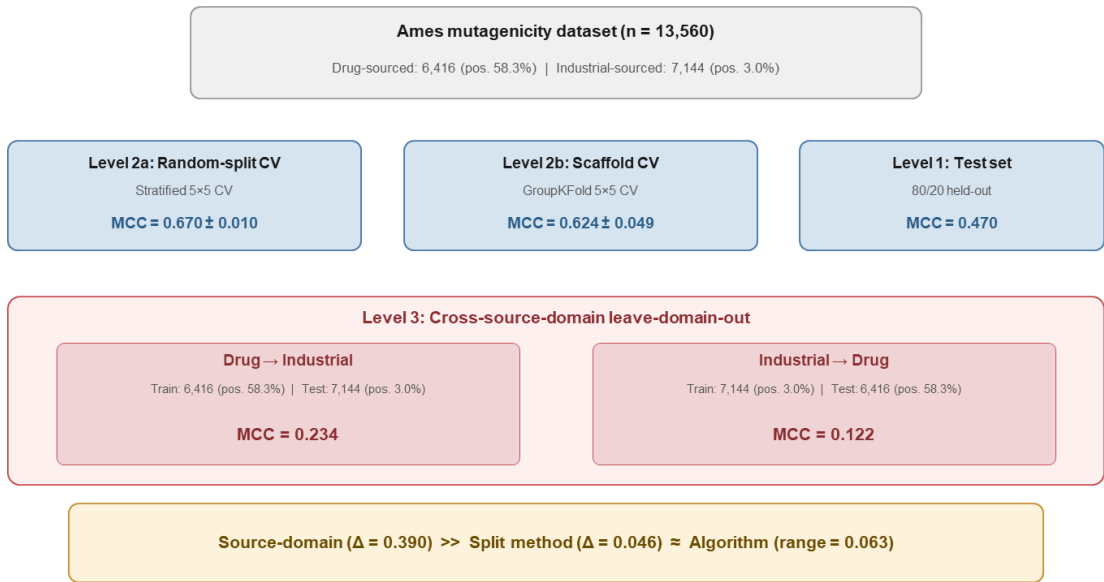


Figure 1: Three-level evaluation framework for genotoxicity QSAR. All three levels use the same locked configuration (XGBoost, compact features, raw SMILES) to enable fair comparison of MCC across evaluation methods. The overestimation hierarchy summarizes the relative magnitudes observed in this benchmark.

Results

Best Observed Performance Across the Full Grid

Table 2 presents the best observed performance for each primary endpoint across all 375 configurations evaluated on the scaffold-split test set. These values represent an upper bound of achievable performance under extensive model/feature/preprocessing search and should be interpreted as a descriptive benchmark, not as unbiased generalization estimates. The confirmatory analysis uses the locked configuration (Table 4).

Table 2: Best observed performance per primary endpoint (scaffold-split test set, full grid). These represent best performance among all 375 evaluated configurations and should be interpreted as a descriptive benchmark, not unbiased generalization estimates. Bootstrap CIs reflect test-set sampling variability but do not account for model-selection variance. Confirmatory estimates from the locked configuration appear in Table 4.

Endpoint	Algo.	Preproc.	Features (n)	MCC [95% CI]	AUC	PR-AUC	Sens.	Spec.
Ames	XGB	raw	FP-2048 (522)	0.555 [.522,.588]	0.850	0.765	0.756	0.822
In vitro	LGBM	salt_str.	FP-1024 (394)	0.297 [.131,.450]	0.665	0.278	0.349	0.929
In vivo	XGB	salt_str.	FP-512 (266)	0.311 [−.014,.571]	0.860	0.237	0.319	0.983

PR-AUC baselines (prevalence): Ames 0.302, in vitro 0.124, in vivo 0.028.

For Ames, XGBoost with broad FP-2048 achieved $\text{MCC} = 0.555$ (95% CI: 0.522–0.588), $\text{ROC-AUC} = 0.850$, and $\text{PR-AUC} = 0.765$. For in vitro CA, LightGBM with salt-stripped broad FP-1024 achieved $\text{MCC} = 0.297$ (95% CI: 0.131–0.450) and $\text{PR-AUC} = 0.278$. For in vivo MN, XGBoost with salt-stripped broad FP-512 achieved $\text{MCC} = 0.311$ (95% CI: −0.014 to 0.571), $\text{ROC-AUC} = 0.860$, and $\text{PR-AUC} = 0.237$ (baseline PR = 0.028).

Under a fixed feature and preprocessing configuration (compact features, raw SMILES), all seven algorithms were compared for Ames on the scaffold-split test set (Table 3). MCC ranged from 0.430 (Logistic Regression) to 0.492 (Random Forest), a span of 0.063 across all algorithms. The five tabular models excluding Logistic Regression spanned only 0.023 (0.470–0.492). GCN achieved 0.445. Pairwise McNemar tests³⁸ between the top two models yielded no significant differences (Ames $p = 0.39$; in vivo $p = 0.72$). However, this in-domain

equivalence did not extend to cross-domain settings (Source Heterogeneity and Algorithm-Specific Robustness).

Table 3: All-algorithm comparison under locked features (compact, raw SMILES, Ames scaffold-split test set).

Algorithm	MCC	ROC-AUC
Random Forest	0.492	0.838
ANN (MLP)	0.489	0.834
LightGBM	0.479	0.833
SVM (RBF)	0.470	0.832
XGBoost	0.470	0.825
GCN	0.445	0.815
Logistic Regression	0.430	0.800

Range across all 7 algorithms: 0.063. Range excluding LR and GCN: 0.023.

The GCN model underperformed fingerprint-based approaches on all endpoints (Ames mean MCC = 0.402 versus 0.507 for XGBoost), consistent with prior observations that graph neural networks require substantially larger datasets to outperform engineered features.

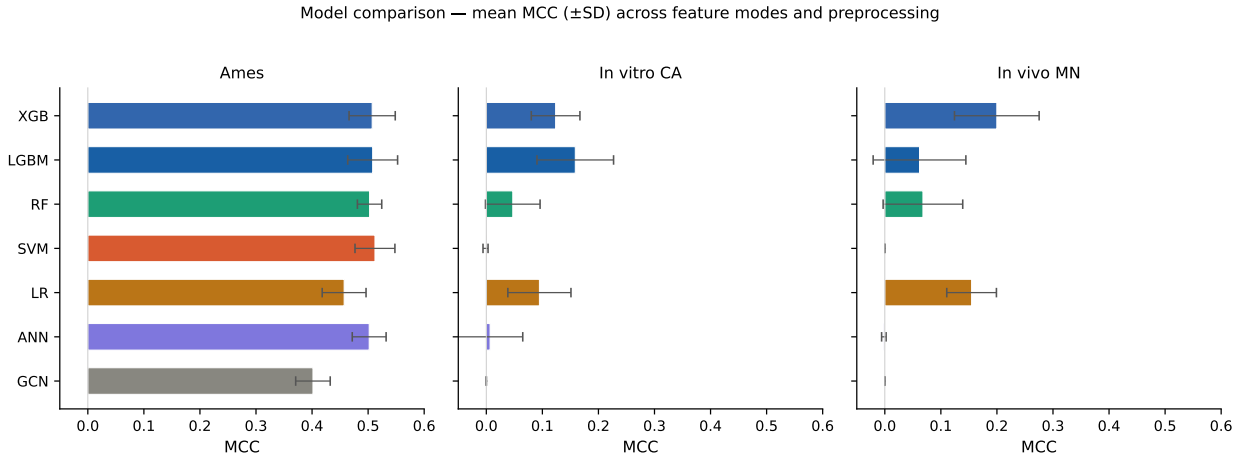


Figure 2: Mean MCC ($\pm$ SD) by algorithm across all feature modes and preprocessing scenarios, grouped by endpoint. Error bars represent ± 1 SD. Within each endpoint, the top five tabular models show minimal performance spread (Ames range = 0.023 excluding LR and GCN).

Evaluation Rigor Cascade (Locked Configuration)

Table 4 and Figure 3 present results exclusively from the locked configuration (XGBoost, compact features, raw SMILES), ensuring that observed differences reflect only the evaluation method.

Table 4: Evaluation rigor cascade—locked configuration (XGBoost, compact features, raw SMILES). Levels ordered 2→1→3 to show progression from within-distribution CV through single-split held-out to cross-domain transfer. All values from the same locked configuration.

Level	Evaluation method	Ames MCC	In vitro MCC	In vivo MCC
2a	Random-split 5×5 CV	0.670 ± 0.010	0.180 ± 0.053	0.181 ± 0.094
2b	Scaffold-split 5×5 CV	0.624 ± 0.049	0.189 ± 0.084	0.104 ± 0.111
1	Scaffold-split test	0.470 [.434,.502]	0.114 [−.023,.270]	0.071 [−.029,.285]
3a	LDO: drug → industrial	0.234	—	—
3b	LDO: industrial → drug	0.122	—	—

Level 2: mean ± SD (25 folds). Level 1: MCC [95% bootstrap CI]. Level 3: Ames only.

For Ames, random-split CV yielded $\text{MCC} = 0.670 \pm 0.010$, consistent with published benchmarks.^{7,10} Scaffold-split CV yielded $\text{MCC} = 0.624 \pm 0.049$, a reduction of 0.046. The scaffold-split held-out test yielded $\text{MCC} = 0.470$ (95% CI: 0.434–0.502).

Cross-source LDO revealed a substantially larger performance collapse. Drug → industrial yielded $\text{MCC} = 0.234$; industrial → drug yielded $\text{MCC} = 0.122$.

Comparing magnitudes: the source-heterogeneity gap (scaffold CV 0.624 → LDO drug→industrial 0.234, $\Delta = 0.390$) substantially exceeded the split-method gap (random CV 0.670 → scaffold CV 0.624, $\Delta = 0.046$). We note that the LDO comparison involves additional factors beyond source membership (directional asymmetry, different test set sizes and class distributions), so $\Delta = 0.390$ should be interpreted as reflecting the combined effect of source shift plus these structural differences. Nevertheless, the magnitude difference between the two effects is unambiguous in this benchmark.

For in vitro CA, random and scaffold splits produced nearly identical results (MCC 0.180 versus 0.189, $\Delta = -0.009$). For in vivo MN, random split produced $\text{MCC} = 0.181$ versus scaffold split $\text{MCC} = 0.104$ ($\Delta = 0.077$), though both values represent near-random

prediction given their standard deviations (0.094 and 0.111).

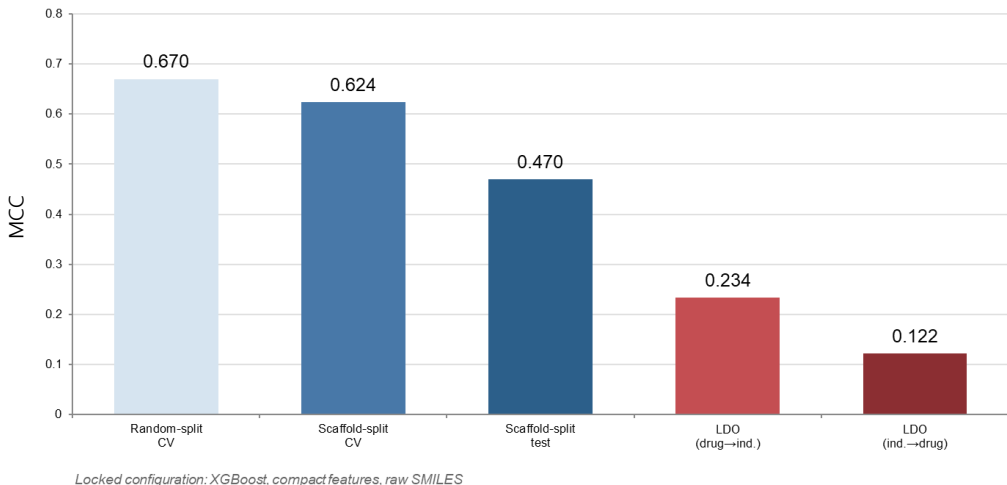


Figure 3: Evaluation rigor cascade for Ames mutagenicity (locked configuration). MCC decreases from random-split CV (0.670) through scaffold-split CV (0.624) and scaffold-split held-out test (0.470) to cross-source LDO (0.234 drug→industrial, 0.122 industrial→drug). The source-heterogeneity effect ($\Delta = 0.390$) substantially exceeds the split-method effect ($\Delta = 0.046$). All values from the same model, features, and preprocessing.

Source Heterogeneity and Algorithm-Specific Robustness

The Ames dataset exhibited severe source heterogeneity (Figure 4). Drug-sourced compounds ($n = 6,416$, positive rate 58.3%) and industrial-sourced compounds ($n = 7,144$, positive rate 3.0%) differed by a factor of 19.5 in mutagenicity prevalence. A naive classifier predicting positive for drug-sourced and negative for industrial-sourced compounds achieved $\text{MCC} = 0.608$ and balanced accuracy = 0.834, comparable to the random-split CV MCC of 0.670, without any structural information.

LDO with three algorithms revealed two findings (Table 5). First, drug $\rightarrow$ industrial generalization was consistent across algorithms (MCC 0.232–0.235), suggesting sufficient positive examples in the drug training set.

Second, industrial $\rightarrow$ drug generalization exposed substantial algorithm-specific differ-

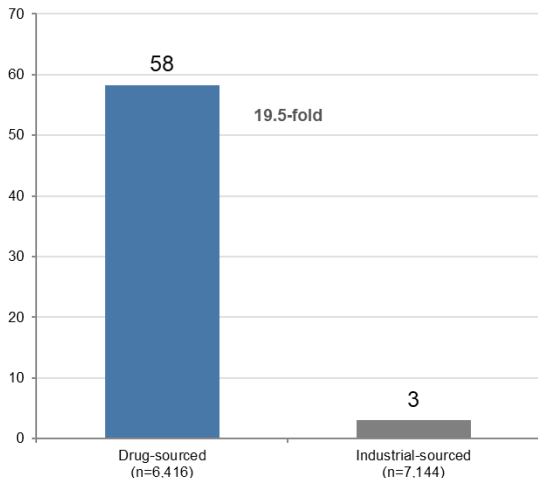
ences. XGBoost achieved $\text{MCC} = 0.122$ (sensitivity 0.062), LightGBM 0.134 (sensitivity 0.081), while Random Forest maintained $\text{MCC} = 0.325$ (sensitivity 0.466). We hypothesize that Random Forest’s superior cross-domain robustness arises from feature bagging,²⁹ which reduces reliance on domain-correlated feature subsets.

Table 5: Leave-domain-out results by algorithm (Ames).

Algorithm	Drug $\rightarrow$ Ind. MCC	Ind. $\rightarrow$ Drug MCC	Ind. $\rightarrow$ Drug Sens.
XGBoost	0.234	0.122	0.062
LightGBM	0.232	0.134	0.081
Random Forest	0.235	0.325	0.466

Drug: $n = 6,416$, pos. rate = 58.3%. Industrial: $n = 7,144$, pos. rate = 3.0%. Prevalence ratio = 19.5-fold.
Naive source classifier: $\text{MCC} = 0.608$, balanced accuracy = 0.834.

(A) Source-domain composition



Naive source-domain classifier: $\text{MCC} = 0.608$, balanced accuracy = 0.834

(B) Leave-domain-out by algorithm

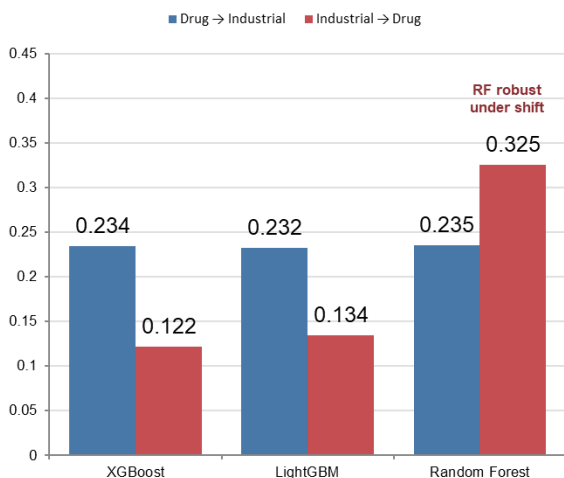


Figure 4: Source heterogeneity and cross-source robustness in the Ames dataset. (A) Positive rate by data source: drug-sourced 58.3% vs. industrial-sourced 3.0% (19.5-fold). (B) Leave-domain-out MCC for three algorithms, showing Random Forest’s substantially better robustness in the industrial $\rightarrow$ drug direction ($\text{MCC} = 0.325$ vs. XGBoost 0.122).

Preprocessing Impact

Salt stripping effects were endpoint-specific (Figure 5). For in vivo MN, the best salt-stripped model achieved $\text{MCC} = 0.311$, substantially outperforming the best raw model ($\text{MCC} = 0.086$), an improvement of $+0.225$. Salt fragments were present in 83% of in vivo SMILES. Among salt-containing in vivo compounds, the positive rate was 2.8%, while non-salt compounds showed 7.1%.

For Ames, salt stripping produced mixed effects ($\Delta = -0.03$ to $+0.02$). For in vitro CA, salt stripping was generally detrimental (mean MCC decreased from 0.081 to 0.056). This endpoint specificity challenges the common practice of applying uniform preprocessing¹⁶ and argues for per-endpoint optimization.

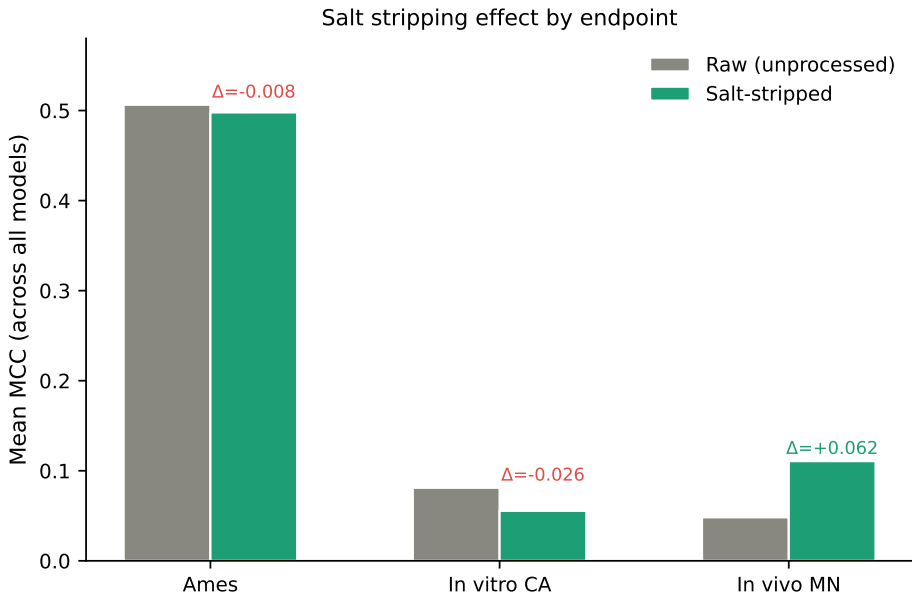


Figure 5: Effect of salt stripping on mean MCC (averaged across all models and feature modes) by endpoint. Salt stripping improved in vivo mean MCC from 0.048 to 0.111 ($\Delta = +0.063$); the best individual in vivo model showed $+0.225$. Ames showed mixed effects; in vitro showed a decrease.

In Vivo Micronucleus: Ranking Signal Versus Classification Utility

The in vivo MN results require careful interpretation. The best model achieved ROC-AUC = 0.860 and PR-AUC = 0.237 (versus baseline PR of 0.028, an 8.5-fold enrichment). However, MCC = 0.311 with 95% CI crossing zero (-0.014 to 0.571) indicates unreliable classification, reflecting extreme class imbalance (training positive rate 3.7%; test set: 12 positives out of 431, 2.8%).

The learning curve (locked configuration) showed flat MCC (≈ 0.055) from 10% to 100% of training data (Figure 7). This curve was not computed for the best-performing configuration, so the flat profile may partly reflect baseline feature and preprocessing choices.

In vivo QSAR may have value as a component of tiered screening³⁹ but should not serve as a standalone binary classifier with current 2D descriptors.

Feature Importance

SHAP analysis³⁷ revealed consistent patterns (Figure 6). Physicochemical descriptors dominated: alert count (mean $|\text{SHAP}| = 0.82$) and MW (0.57) for Ames; LogP (0.58) for in vitro; MW (1.00) for in vivo. Individual fingerprint bits were absent from the top-5 despite comprising 50–75% of features. Compact features yielded mean MCC = 0.448, while broad FP-2048 yielded 0.523 ($\Delta = +0.075$, averaged across all Ames models and preprocessing scenarios).

Learning Curves

Learning curves for the locked configuration (Figure 7) revealed three patterns: Ames MCC saturated early (0.543 at 10% of data, 0.640 at 100%); in vitro showed a noisy upward trend ($0.070 \rightarrow 0.170$); in vivo was flat (≈ 0.055 throughout). As noted in In Vivo Micronucleus: Ranking Signal Versus Classification Utility, the in vivo curve reflects the locked configuration only.

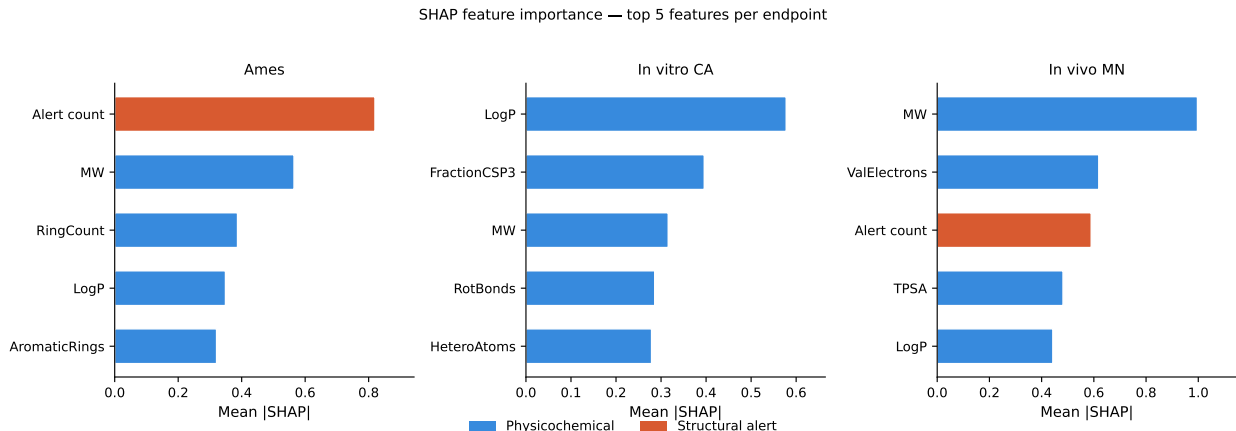


Figure 6: SHAP feature importance (top 5 features) per primary endpoint. Physicochemical descriptors (blue) dominate; structural alert count (orange) ranks first for Ames. Individual fingerprint bits are absent from the top 5 despite comprising 50–75% of features.

Discussion

The Performance Variation Hierarchy in This Benchmark

Using a locked model configuration, we observed the following hierarchy of effects on apparent Ames QSAR performance:

Source heterogeneity $\gg$ Data splitting method $\approx$ In-domain algorithm choice

The source-heterogeneity gap (scaffold CV $\rightarrow$ LDO, $\Delta = 0.390$) substantially exceeded the split-method gap (random $\rightarrow$ scaffold CV, $\Delta = 0.046$), which was comparable to the in-domain algorithm spread (MCC range = 0.063 across all seven algorithms under fixed features and preprocessing).

Three caveats apply. First, this hierarchy was established on a single Ames benchmark where source labels were available; other datasets may exhibit different heterogeneity patterns. Second, the LDO $\Delta = 0.390$ reflects the combined effect of source shift plus structural differences (asymmetric test sizes, different class distributions per source) and should be interpreted as an upper bound. Third, algorithm robustness was not equivalent under source shift (Source Heterogeneity and Algorithm-Specific Robustness) so algorithm choice

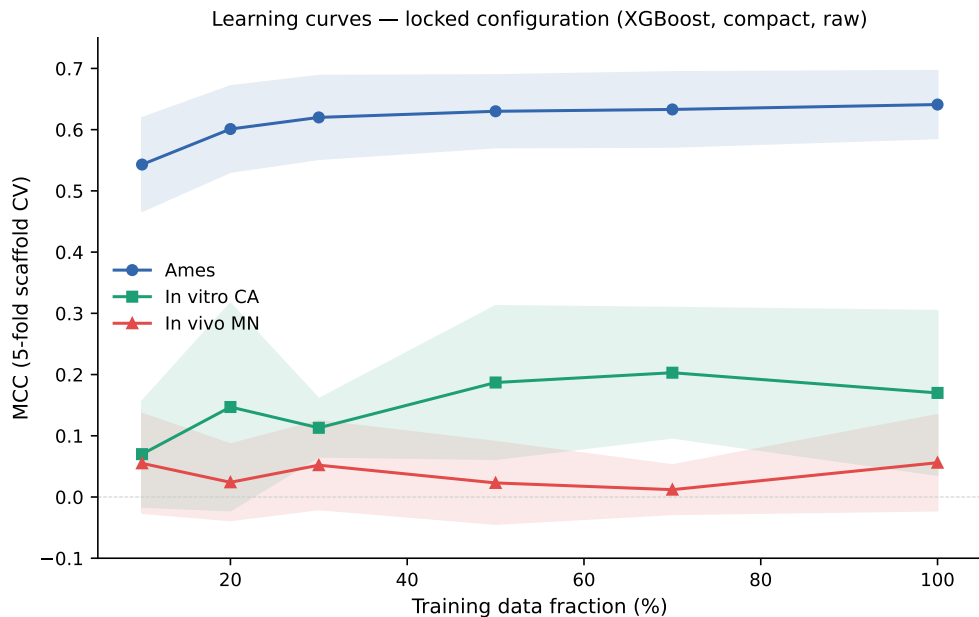


Figure 7: Learning curves (locked configuration). Ames MCC saturates by 30% of data; in vitro shows a noisy upward trend; in vivo is flat. Shaded regions: ± 1 SD across 5-fold scaffold CV.

may matter substantially more in cross-source deployment than the in-domain comparison suggests.

Possible Drivers of Source Heterogeneity

A key question is what drives the observed performance gap between drug-sourced and industrial-sourced compounds. We identify at least three non-exclusive hypotheses:

Chemical space difference. Pharmaceutical and industrial chemical libraries may occupy different regions of chemical property space.^{40,41} Drug candidates are designed for biological activity and tend to follow Lipinski-like property distributions, while industrial chemicals span a broader structural diversity. The 19.5-fold prevalence difference is consistent with—though not proof of—distinct chemical domains.

Data quality and labeling heterogeneity. The two sources differ in provenance: drug-sourced compounds originate from curated pharmaceutical benchmark datasets, while industrial compounds are collected from regulatory submissions (e.g., ECHA, K-REACH)

where testing protocols, laboratory standards, and reporting conventions may vary. Differences in assay sensitivity, strain selection, or dose ranges across laboratories could contribute to the observed prevalence gap independently of genuine chemical differences.

Selection bias. Drug candidates are intentionally designed to interact with biological targets, increasing the likelihood of genotoxic activity as a side effect. Industrial chemicals are selected for manufacturability and stability, not biological activity, which may inherently bias them toward lower mutagenicity rates. This is a sampling property, not a chemical domain difference per se.

Our current experimental design cannot distinguish between these hypotheses. A matched-subset analysis controlling for chemical similarity across sources, chemical space visualization (e.g., UMAP on molecular descriptors⁴²), and source-stratified performance assessment would help disentangle these factors and represent priorities for future work. Regardless of the underlying cause, the practical implication is the same: QSAR models trained on mixed-source data should be evaluated for cross-source generalization before deployment.

In Vivo Prediction: Implications for Tiered Testing

The dissociation between ranking ability ($\text{ROC-AUC} = 0.860$, $\text{PR-AUC} = 0.237$) and classification reliability (MCC CI crossing zero) has practical implications. For ICH M7-style binary hazard classification,¹ in vivo MN QSAR does not achieve sufficient reliability with current 2D descriptors. However, the PR-AUC of 0.237 represents an 8.5-fold enrichment over baseline prevalence (2.8%), suggesting potential value in tiered screening strategies.³⁹

The strong preprocessing sensitivity (+0.22 MCC from salt stripping) indicates that reported in vivo QSAR performance depends critically on preprocessing choices that are often undocumented.¹⁷ The biological plausibility of in vivo prediction limitations is high: micronucleus induction depends on ADME properties, metabolic activation, tissue distribution, and DNA repair capacity,^{43,44} none of which are captured by 2D descriptors.

Cross-Endpoint Overlap

The high compound overlap between endpoints ($>90\%$ shared with Ames) indicates that performance differences across endpoints (Ames MCC = 0.555 versus in vivo MCC = 0.311) are unlikely to be primarily explained by different chemical spaces alone, but rather reflect inherent differences in endpoint predictability from 2D structure.¹⁴ Multi-task learning approaches jointly predicting all endpoints may be a promising direction for future work.⁴⁵

Preprocessing Deserves Systematic Treatment

Salt stripping effects were endpoint-specific: beneficial for in vivo (+0.22 best-model improvement), mixed for Ames (± 0.03), and generally detrimental for in vitro (-0.03 to -0.13). We hypothesize that salt stripping harms in vitro CA performance because some counterions are themselves relevant to the clastogenicity assay. A stratified analysis by salt-containing versus non-salt and metal versus non-metal compounds would directly test this hypothesis and represents a tractable extension. The QSAR data curation literature^{16,17} has emphasized the importance of standardization, but our results suggest that the optimal standardization strategy is endpoint-dependent.

Practical Recommendations

Based on our findings, we propose the following reporting standards for genotoxicity QSAR:

1. Use scaffold-based splitting for primary performance reporting.
2. When source labels are available, report leave-domain-out performance and disclose source composition with per-source positive rates.
3. Optimize preprocessing per endpoint rather than applying uniform pipelines.
4. Report bootstrap CIs and applicability domain coverage.

5. For endpoints with extreme class imbalance, separately report threshold-free metrics (ROC-AUC, PR-AUC) and threshold-dependent metrics (MCC, balanced accuracy).
6. When claiming algorithm superiority, test under domain shift as well as in-domain conditions.

Limitations

(a) Source heterogeneity analysis was limited to Ames; other endpoints lacked source annotations. (b) Source labels reflect data provenance rather than computed chemical classification; the relative contributions of chemical space difference, data quality variation, and selection bias to the observed performance gap remain unresolved. (c) The in vivo learning curve used only the locked configuration, not the best-performing configuration. (d) A single scaffold split was used rather than repeated holdout. (e) The GCN did not include edge features or pre-training.⁴⁶ (f) No 3D descriptors, QM descriptors, or metabolite predictions were included. (g) Within-AD versus outside-AD performance stratification for LDO was not conducted. (h) McNemar tests were limited to pairwise top-two comparisons; comprehensive multiple-comparison correction²⁰ was not applied. (i) Table 2 reports best observed performance across 375 configurations evaluated on the test set, which introduces model-selection bias; the locked-configuration results (Table 4) provide unbiased estimates.

Conclusion

We present a multi-level evaluation of genotoxicity QSAR revealing that, for the Ames mutagenicity benchmark studied here, source heterogeneity—arising from the mixing of drug-like and industrial chemicals with distinct provenance, prevalence, and potentially different data quality—drove a substantially larger performance gap ($\Delta = 0.390$ from scaffold CV to LDO) than the choice between random and scaffold splitting ($\Delta = 0.046$). Algorithm differences were small in the in-domain setting (MCC range = 0.063) but emerged under source shift,

with Random Forest showing better cross-source robustness than gradient boosting. In vivo micronucleus models showed ranking signal but lacked reliable classification utility with current 2D descriptors. We recommend multi-level evaluation with source-awareness as a minimum reporting standard for Ames mutagenicity QSAR.

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Supporting Information

The Supporting Information is available free of charge. Summary tables are shown below; complete per-configuration data are provided as supplementary CSV files in the code repository.

Table S1. Sampling endpoint results

Table S1: Best model per sampling endpoint (exploratory). Small test sets; interpret with caution.

Endpoint	Best model	MCC [95% CI]	AUC	Sens.	n_{test}
In vitro (sampled)	RF / FP-512	0.663 [.459,.839]	0.827	0.714	69
In vivo (sampled)	XGB / FP-1024	0.832 [.644,1.00]	0.946	0.750	47

Table S2. Random versus scaffold split comparison

Table S2: Random vs. scaffold split (XGBoost, compact features, raw SMILES, 5×5 CV).

Endpoint	n	Random MCC	Scaffold MCC	Δ
Ames	13,560	0.670 ± 0.010	0.624 ± 0.049	0.046
In vitro CA	1,619	0.180 ± 0.053	0.189 ± 0.084	-0.009
In vivo MN	2,153	0.181 ± 0.094	0.104 ± 0.111	0.077
In vitro (sampled)	345	0.393 ± 0.085	0.367 ± 0.085	0.026
In vivo (sampled)	231	0.525 ± 0.152	0.429 ± 0.157	0.096

Table S3. Full 375-experiment results

Complete metrics (MCC, bootstrap CI, ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, Brier score, AD coverage) for all 375 configurations are provided in `table_s3_full_results.csv`

Table S4. Hyperparameter tuning comparison

Default versus tuned MCC for all configurations, including optimized parameter values, are provided in `table_s4_hp_tuning.csv`.

Table S5. Applicability domain coverage

Per-configuration AD coverage (Tanimoto ≥ 0.3 , 1024-bit Morgan FP) and mean similarity values are provided in `table_s5_ad_coverage.csv`. Coverage ranged from 48.5% (in vitro) to 58.9% (in vivo) across configurations.

Table S6. Cross-endpoint compound overlap

Pairwise compound overlap and label concordance (κ) between all endpoint datasets are provided in `table_s6_cross_endpoint.csv`. Overlap with Ames was 97.5% for in vitro and 92.1% for in vivo.

Table S7. SHAP feature importance

SHAP values (mean $|\text{SHAP}|$, TreeExplainer) for all features across all tree-based models are provided in `table_s7_shap.csv`.

TOC Graphic

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